

Information System Audit (Elective)

Course Code: ISA02022

Lecture : (4 hr. Per Week)

Year: II

Part: I

Course Objectives

This course covers different concepts of Information Systems Auditing including basics, hardware and software security issues, information systems audit and conducting IS audit, risk-based systems audit, business continuity and disaster, auditing in the ICT environment, and security testing.

- To explore concept information systems auditing
- To know hardware and software security issues
- To understand and conducting information systems audit
- To know about risk based systems audit
- To study business continuity and disaster recovery concepts
- To understand auditing in ICT environments
- To know about security testing tools

Course Contents

Unit 1: Overview of Systems Audit (10 Hrs.)

- 1.1. Information Systems Audit and Information Systems Auditor
- 1.2. Legal Requirements of an Information Systems Audit
- 1.3. Systems Environment and Information Systems Audit
- 1.4. Information Systems Assets and Classification of Controls
- 1.5. Impact of Computers on Auditing
- 1.6. Information Systems Audit Coverage
- 1.7. IT Audit Framework, ISO 27001:2013, NIST Cyber Security Framework, COBIT, CIS

Unit 2: Hardware and Software Security Issues (10 Hrs.)

- 2.1. Hardware Security Objective
- 2.2. Peripheral Devices and Storage Media
- 2.3. Authentication Devices
- 2.4. Hardware Acquisition, Hardware Maintenance and Management of Obsolescence
- 2.5. Disposal of Equipment; Problem Management; Change Management
- 2.6. Network and Communication Issues.
- 2.7. Overview of Types of Software; Elements of Software Security
- 2.8. Control Issues during Installation and Maintenance
- 2.9. Licensing Issues

Unit 3: Information Systems Audit Requirements (10 Hrs.)

- 3.1. Risk Analysis; Threats, Vulnerability, Exposure, Likelihood, and Attack
- 3.2. Information Systems Control Objectives; Information Systems Audit Objectives;
- 3.3. System Effectiveness and Efficiency
- 3.4. Information Systems Abuse
- 3.5. Asset Safeguarding Objective and Process
- 3.6. Evidence Collection and Evaluation
- 3.7. Logs and Audit Trails as Evidence

Unit 4: Conducting an Information Systems Audit (12 Hrs.)

- 4.1. Audit Program and Audit Plan
- 4.2. Audit Procedures and Approaches
- 4.3. System Understanding and Review
- 4.4. Compliance Reviews and Tests
- 4.5. Substantive Reviews and Tests
- 4.6. Audit Tools and Techniques
- 4.7. Sampling Techniques
- 4.8. Audit Questionnaire; Audit Documentation; Audit Report
- 4.9. Auditing Approaches; Sample Audit Work-Planning Memo
- 4.10. Sample Audit Work Process Flow
- 4.11. Conducting a Risk-Based Information Systems Audit
- 4.12. Risk Assessment and Risk Management Strategy.

Unit 5: Business Continuity and Disaster Recovery Plan (10 Hrs.)

- 5.1. Business Continuity and Disaster Recovery Process
- 5.2. Business Impact Analysis; Incident Response Plan
- 5.3. Disaster Recovery Plan
- 5.4. Types of Disaster Recovery Plans
- 5.5. Emergency Preparedness Audit Checklist
- 5.6. Business Continuity Strategies
- 5.7. Business Resumption Plan Audit Checklist
- 5.8. Recovery Procedures Testing Checklist; Plan Maintenance Checklist;

Unit 6: Security Testing and Cloud Computing Audit (8 Hrs.)

- 6.1. Cybersecurity
- 6.2. Global Cybersecurity Landscape
- 6.3. Vulnerability Assessment and Penetration Testing (VAPT)
- 6.4. Secured Software Development Testing,
- 6.5. Open Web Application Security Project
- 6.6. Security Testing Tools
- 6.7. Cloud Audit Considerations

Assignments

1. Information Security Gap Assessment through ISO 27001:2013
2. Cyber Security Maturity Assessment thorough NIST Cyber Security Framework.
3. IT Governance Audit through Control Objective for Related Technologies.
4. Develop IS Audit Terms of Reference
5. IS Audit Case Study

Evaluation Schemes**a. Internal Examination**

Type	Weightage
Minor tests	60%
Assignments	40%

b. Final Examination

There will be five units of questions carrying 12 marks each. The question will cover all chapters of the syllabus. The evaluation scheme will be as indicated in the table:

Units	Chapters	Marks
1	1.1, 1.2, 1.3, 1.4, 1.5. 1.6,1.7,6.7	12
2	2.1, 2.2, 2.3,2.4.2.5, 2.6, 2.7,2.8,2.9	12
3	3.1, 3.2, 3.3, 3.4,3.5, 3.6,3.7,6.4,6.5,6.6	12
4	4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 4.10, 4.11, 4.12	12
5	5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 6.1, 6.2, 6.3	12
Total		60

References

1. Veena Hingarh and Arif Ahmed, *Understanding and Conducting Information Systems Auditing*, Wiley, 2013
2. Jack J. Champlain, *Auditing Information Systems*, 2nd Edition, Wiley, 2003
3. Richard Cascarino, *Auditor's Guide to Information Systems Auditing*, Wiley, 2007
4. ISACA Journals
5. ISACA IT Audit Frameworks
6. NIST Special Publications